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Genome Analysis of the Ancient Tracheophyte *Selaginella tamariscina* Reveals Evolutionary Features Relevant to the Acquisition of Desiccation Tolerance

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ABSTRACT

Resurrection plants, which are the "gifts" of natural evolution, are ideal models for studying the genetic basis of plant desiccation tolerance. Here, we report a high-quality genome assembly of 301 Mb for the diploid spike moss *Selaginella tamariscina*, a primitive vascular resurrection plant. We predicated 27 761 protein-coding genes from the assembled *S. tamariscina* genome, 11.38% (2363) of which showed significant expression changes in response to desiccation. Approximately 60.58% of the *S. tamariscina* genome was annotated as repetitive DNA, which is an almost 2-fold increase of that in the genome of desiccation-sensitive *Selaginella moellendorffii*. Genomic and transcriptomic analyses highlight the unique evolution and complex regulations of the desiccation response in *S. tamariscina*, including species-specific expansion of the oleosin and pentatricopeptide repeat gene families, unique genes and pathways for reactive oxygen species generation and scavenging, and enhanced abscisic acid (ABA) biosynthesis and potentially distinct regulation of ABA signaling and response. Comparative analysis of chloroplast genomes of several *Selaginella* species revealed a unique structural rearrangement and the complete loss of chloroplast NAD(P)H dehydrogenase (NDH) genes in *S. tamariscina*, suggesting a link between the absence of the NDH complex and desiccation tolerance. Taken together, our comparative genomic and transcriptomic analyses reveal common and species-specific desiccation tolerance strategies in *S. tamariscina*, providing significant insights into the desiccation tolerance mechanism and the evolution of resurrection plants.

Key words: resurrection plant, *Selaginella tamariscina*, herbgenomics, evolution, synthetic biology

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INTRODUCTION

Resurrection plants, which include many ferns and mosses and a few angiosperm plants, possess desiccation tolerance (DT) and can endure extreme conditions of only 1%–5% relative water content (Giarola et al., 2017). Although DT is common in seeds of angiosperm plants, approximately 330 resurrection plants have acquired DT in vegetative tissues (Porembski and Barthlott, 2000; Leprince and Buitink, 2015; Costa et al., 2016). DT existed early in land plants as they colonized terrestrial

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habitats surrounding freshwater sources, and was first found in reproductive propagules and then in vegetative tissues (Gaff and Oliver, 2013). During the evolution of land plants, DT in vegetative organs was lost in favor of other developmental traits, such as increased growth rate, and structural and morphological complexity (Oliver et al., 2000; Farrant and Moore, 2011). The reacquisition of DT by vegetative tissues during long-term drought stress may have evolved through reactivation of this innate DT mechanism (Tautz and Domazet-Loso, 2011). The genomes of a few resurrection plants have been sequenced, including the monocots *Oropetium thomaeum* (VanBuren et al., 2015) and *Xerophyta viscosa* (Costa et al., 2017), the dicot *Boea hygrometrica* (Xiao et al., 2015), and the lycophyte *Selaginella lepidophylla* (VanBuren et al., 2018), providing the research community with the valuable resources for studying the molecular basis of DT.

The earliest record of a resurrection plant in China is the description of *Selaginella tamariscina* in the Compendium of Materia Medica, in which this plant is named "long life," "undead," and "whorlleaf stonecrop herb" because of its resurrection ability and pharmacological value. *S. tamariscina* was recorded in the Chinese Pharmacopoeia because of its medicinal value in promoting blood circulation (Chinese Pharmacopoeia Committee, 2015). *Selaginella* is the sole genus in the spike moss family (Selaginellaceae) of lycophytes that belong to the more ancient tracheophytes, which were established in the late Silurian or early Devonian, approximately 400 million years ago (MYA) (Banks, 2009). Phylogenetic relationships estimated via chloroplast or mitochondrial DNA comparisons have shown that the lycophytes are a monophyletic group including the Lycopodiaceae, Isoetaceae, and Selaginellaceae; these families share some common traits such as microphylls and adaxial, reniform sporangia (Raubeson and Jansen, 1992; Duff and Nickrent, 1999). The genus *Selaginella* has been described as an enigmatic, ancient, and distinctive group, containing approximately 700 species that are spread over temperate, tropical, arctic, and desert regions (Banks, 2009). In the Flora of China, 66 *Selaginella* species were recorded; among them, *S. tamariscina* and *Selaginella pulvinata* are listed in the Chinese Pharmacopoeia as medicinal plants (Chen et al., 2015). Based on phylogenetic analyses, the rates of molecular evolution between *Selaginella* species are remarkably higher than those observed for the species in angiosperm families (Korall and Kenrick, 2002, 2004). Since the Permian–Triassic extinction, many *Selaginella* species, such as *S. tamariscina* (Wang et al., 2010), *Selaginella lepidophylla* (Yobi et al., 2013; Pampurova et al., 2014; Rafsanjani et al., 2015), and *Selaginella bryopteris* (Deeba et al., 2016), have evolved DT under drought, but most *Selaginella* species, such as *Selaginella moellendorffii*, lack this trait (Yobi et al., 2012).

The *S. moellendorffii* and *S. lepidophylla* genomes have been sequenced, and the *Selaginella* genus has become a model for evolutionary studies based on its important position in the early evolution of vascular plants (Banks et al., 2011; VanBuren et al., 2018). *Selaginella* species possess relatively small genomes and little variation in genome size, which ranges from 81 to 182 Mb based on flow cytometry (Baniaga et al., 2016). Whole-genome duplication (WGD) is an important evolutionary force in vascular land plants; however, the Selaginellaceae family is the only clade of vascular plants that was proposed to have not undergone a WGD event (Jiao et al., 2011). Comparative metabolic profiling of desiccation-sensitive (*S. moellendorffii*) and desiccation-tolerant (*S. lepidophylla*) species of *Selaginella* revealed that under drought conditions, *S. lepidophylla* maintains significantly higher levels of many sugars, amino acids, and other metabolites, such as sucrose, monosaccharides, polysaccharides, sugar alcohols, the osmoprotectant betaine, flavonoids, γ-glutamyl amino acids, and lipoxygenase. These compounds may help to detoxify reactive oxygen species (ROS) and contribute to DT (Yobi et al., 2012). However, the molecular mechanism that differentiates desiccation-sensitive from desiccation-tolerant *Selaginella* species still remains largely unclear.

In this context, a genomic study of the desiccation-tolerant *S. tamariscina* may provide new insights into the evolution of vascular plants and how resurrection plants acquired the DT. Since high heterozygosity (>1.5%) was previously detected in the *S. moellendorffii* genome (Banks et al., 2011), to sequence the whole genome of *S. tamariscina* we used the PacBio Single Molecule, Real-Time (SMRT) sequencing technology, which generates long reads and has been proved to be more efficient in unraveling complex and highly similar repeat sequences, especially in large, polyploid plant genomes. We obtained a 301-Mb high-quality assembly of *S. tamariscina* genome and analyzed gene family expansion and differential gene expression related to DT through comparative genomic and transcriptomic analyses.

RESULTS

Genome Analysis of the Resurrection Plant *S. tamariscina*

We performed whole-genome SMRT sequencing of the resurrection plant *S. tamariscina* (Figure 1; Supplemental Figures 1 and 2). There was substantial DNA polymorphism (approximately 2% heterozygosity) within the *S. tamariscina* genome (Supplemental Figure 3). The draft genome we assembled consists of 1391 scaffolds spanning 301 Mb with 27 761 protein-coding genes identified through *ab initio* annotation (Supplemental Results and Supplemental Table 1). The GC content of the desiccation-tolerant *S. tamariscina* genome is 37.44%, which is lower than the 44.40% GC content of the genome of the desiccation-sensitive *S. moellendorffii* (Supplemental Table 1).

A total of 250 044 simple sequence repeats were annotated, which would provide valuable molecular markers for future genetic diversity studies of *S. tamariscina* (Supplemental Tables 2–4). The number of microRNA (miRNA) loci identified in *S. tamariscina* (146) is slightly lower than that in *Xerophyta viscosa* (165) and *Boea hygrometrica* (196), but more than 2.5-fold the number of miRNAs in *S. moellendorffii* (58) (Supplemental Table 5). Approximately 60.58% (182 184 894 bp) of the *S. tamariscina* genome was annotated as repetitive DNA (Figure 2A and Supplemental Table 6), which is remarkably higher than the repetitive element content of other resurrection plant genomes, including *Oropetium thomaeum* (43.80%) and *X. viscosa* (18.00%), but lower than that of the *B. hygrometrica* genome (75.75%). Furthermore, the genomic proportion of repetitive DNA is almost 2-fold higher than that in the desiccation-sensitive *S. moellendorffii* genome (37.50%).

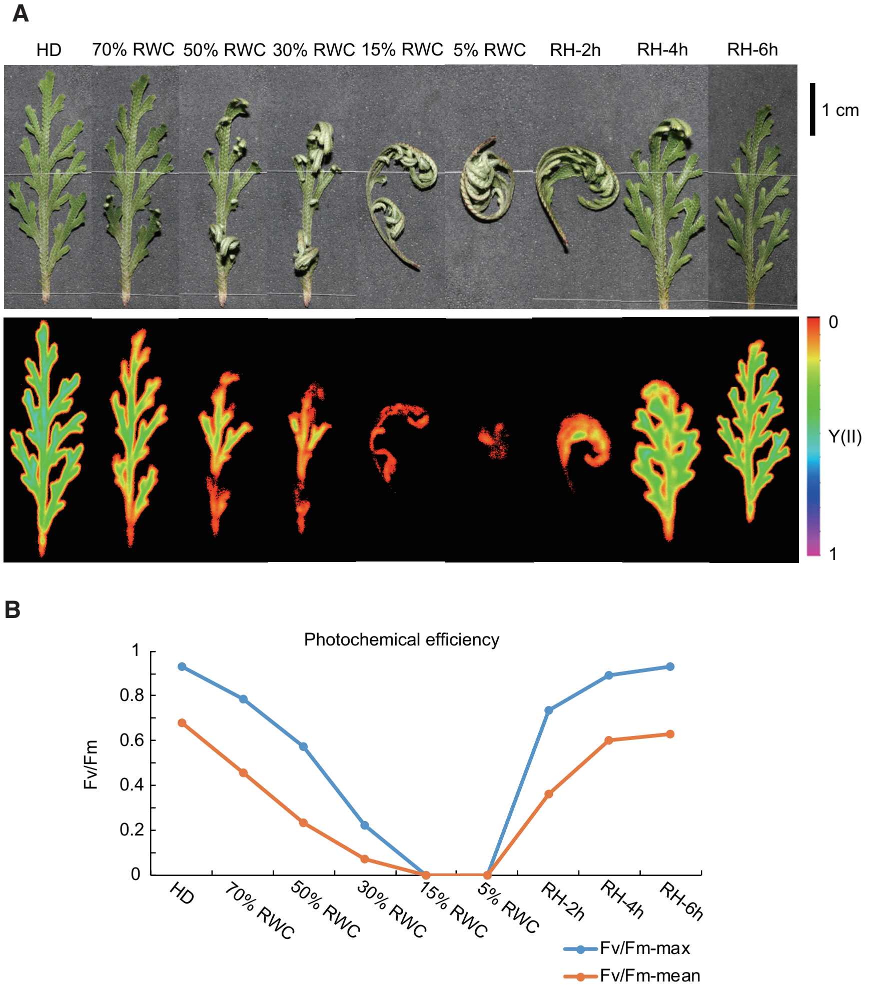

Figure 1. Phenotype, Chlorophyll Fluorescence, and Photochemical Efficiency of *S. tamariscina* during Drought and Rehydration.

(A) Chlorophyll fluorescence Y(II) measured using the MAXI version of the IMAGING-PAM M-Series fluorometer under HD (hydration, 1.49 g H_2O g^{-1} dry weight [dwt]), 70% RWC (1.04 g H_2O g^{-1} dwt), 50% RWC (0.75 g H_2O g^{-1} dwt), 30% RWC (0.46 g H_2O g^{-1} dwt), 15% RWC (0.22 g H_2O g^{-1} dwt), 5% RWC (0.07 g H_2O g^{-1} dwt), RH 2 h (rehydration, 1 g H_2O g^{-1} dwt), RH 4 h (1.28 g H_2O g^{-1} dwt), and RH 6 h (1.49 g H_2O g^{-1} dwt).

(B) The photochemical efficiency (Fv/Fm) of *S. tamariscina* during dehydration and rehydration. The maximum and mean Fv/Fm values represent the maximal and mean photochemical efficiencies, respectively.

Long terminal repeat retrotransposons (LTR-RTs) make up a high proportion (25.47%) of the repetitive elements in *S. tamariscina*; 1277 intact LTR-RTs, including 1059 *Gypsy* and 30 *Copia* elements, were identified. The intact LTR-RTs in *S. tamariscina* were estimated to have an average insertion time of 1.35 MYA, which is much older than the average of 0.37 MYA reported for *S. lepidophylla*, and slightly older than the average insertion times calculated for *Oryza sativa* (0.92 MYA) and *Arabidopsis thaliana* (0.94 MYA) based on a mutation rate of $\mu = 1.3 \times 10^{-8}$ (per bp per year) (VanBuren et al., 2018) (Supplemental Figure 6 and Supplemental Table 7).

Gene Family Evolution

The protein sequences of 18 vascular plants and non-vascular plants were clustered (Nordberg et al., 2014), yielding a total of 48 914 orthologous groups that cover 449 111 genes. From these orthologous groups, 61 589 genes clustering into 1808 gene families were found in all 18 plants, and represent evolutionarily conserved ancestral gene families (Figure 2B). In *S. tamariscina*, 3568 genes from 1345 gene families appeared to be unique. Functional annotation of these *S. tamariscina* genes revealed that 331 are pentatricopeptide repeat (PPR; PF01535) genes, which play broad and essential roles in plant growth and development and represent one of the largest gene families.

Gene family evolution, including gene expansion and contraction, among 18 plant species was further analyzed with CAFÉ. We found that 2620 gene families expanded and 3927 gene families contracted in the *S. tamariscina* lineage (Figure 2C and Supplemental Figure 7). Based on phylogenomic analysis of single-copy genes, *S. tamariscina* and *S. moellendorffii* grouped into one branch, with an estimated divergence time of approximately 295 MYA (Figure 2C).

Structural Loss and Rearrangement within the Chloroplast Genome

The 126 399-bp complete chloroplast genome of *S. tamariscina* was assembled via two different *de novo* assembly strategies using PacBio reads and Illumina reads, respectively. The *S. tamariscina* chloroplast genome contains four regions, namely, large single-copy (LSC; 53 176 bp), small single-copy (SSC; 47 573 bp), and repeats A and B (RA, RB; 12 825 bp), and 55 protein-coding genes were annotated (Figure 3). Almost all chloroplast genomes in the plant kingdom sequenced to date have two conserved inverted repeat (IR) regions (IRA and IRB); however, in the *S. tamariscina* chloroplast genome, two repeat regions are positioned between LSC and SSC. The chloroplast genome assembly was further confirmed by PCR amplification and Sanger sequencing (Supplemental Figure 8 and Supplemental Table 8). Compared with other reported *Selaginella* chloroplast genomes, a structural rearrangement of large fragments in *S. tamariscina* was also observed. The chloroplast genome can be divided into 10 large fragments. Among these, the fragments 2, 3, 4, 5, and 6 in *S. tamariscina* showed an inverted alignment with the corresponding *S. moellendorffii* fragments. In addition, the fragments 1, 7, 8, and 10 underwent inverted or irregular rearrangements during evolution. Compared with the *S. moellendorffii* chloroplast genome, 11 chloroplast NAD(P)H dehydrogenase genes (*cp-ndhA-K*) were found to be completely absent in *S. tamariscina*, and two protochlorophyllide reductase subunits (*chlB* and *chlN*), six ribosomal protein-coding genes (*rpl2*, *rpl16*, *rpl20*, *rpl33*, *rpl36*,

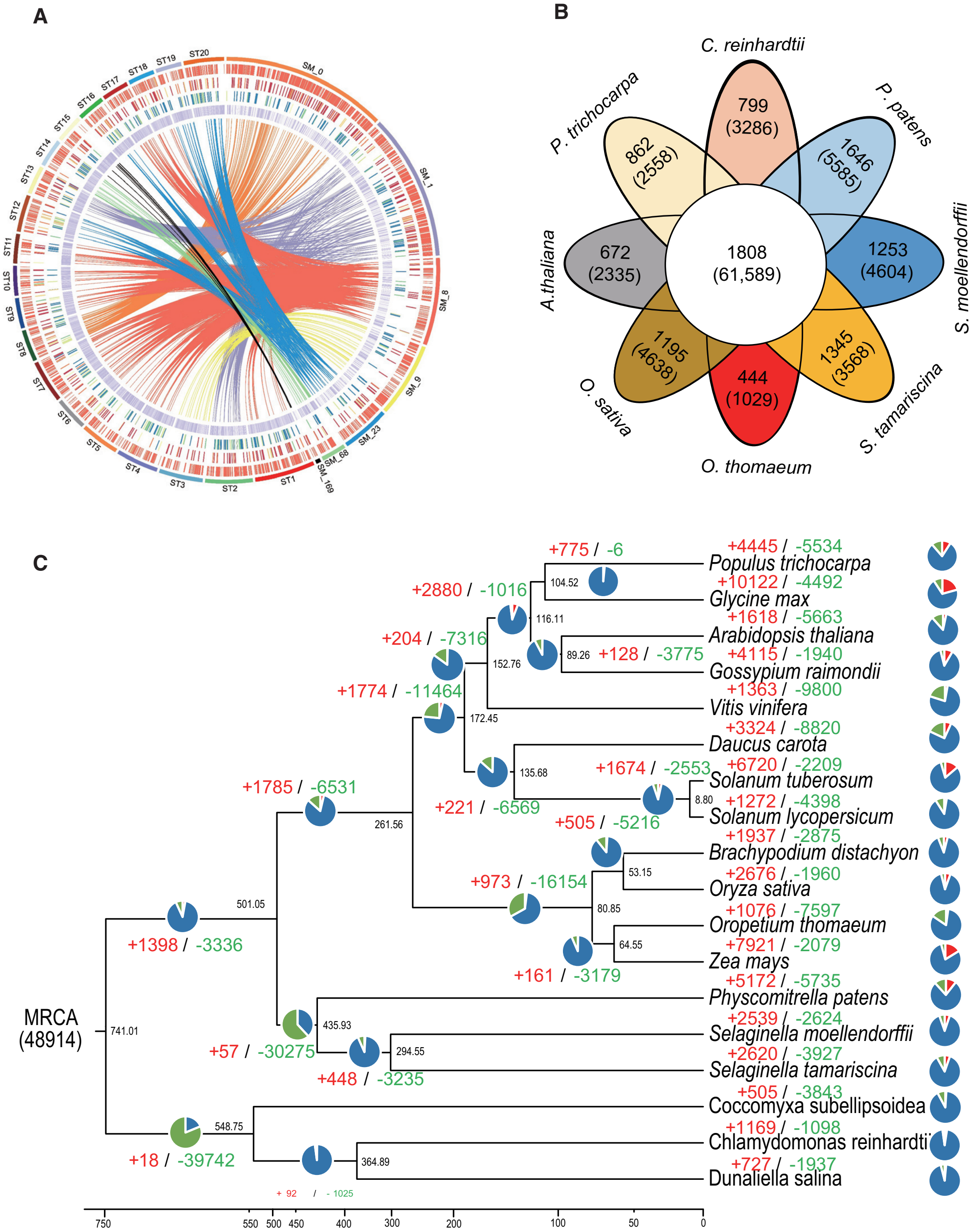

Figure 2. Evolutionary Analysis of the *S. tamariscina* Genome.

(A) An ideogram showing the features of the 20y longest scaffolds from the *S. tamariscina* genome. The circus map shows, from outside to inside, the gene density, upregulated differentially expressed genes (DEGs), downregulated DEGS, transposable element density, and sequence similarity between *S. tamariscina* and *S. moellendorffii*.

(B) Venn diagram illustrating the gene families and genes unique to *S. tamariscina* and those common to seven other plant genomes, including chlorophytes, mosses, lycophytes, and euphyllophytes.

(C) Expansion and contraction of gene families among 18 plant species. Phylogenetic analysis of these 18 species was performed with 18 high-quality 1:1 single-copy orthologous genes. Divergence time was estimated based on the divergence time of *P. trichocarpa*–*G. max* (94–127 MYA), *A. thaliana*–*G. raimondii* (81–94 MYA), and *S. tuberosum*–*S. lycopersicum* (5.09–10.25 MYA). Green and red numbers indicate gene family contraction and expansion events, respectively. The pie charts show the proportion of expanded/contracted/remained gene families in each plant species.

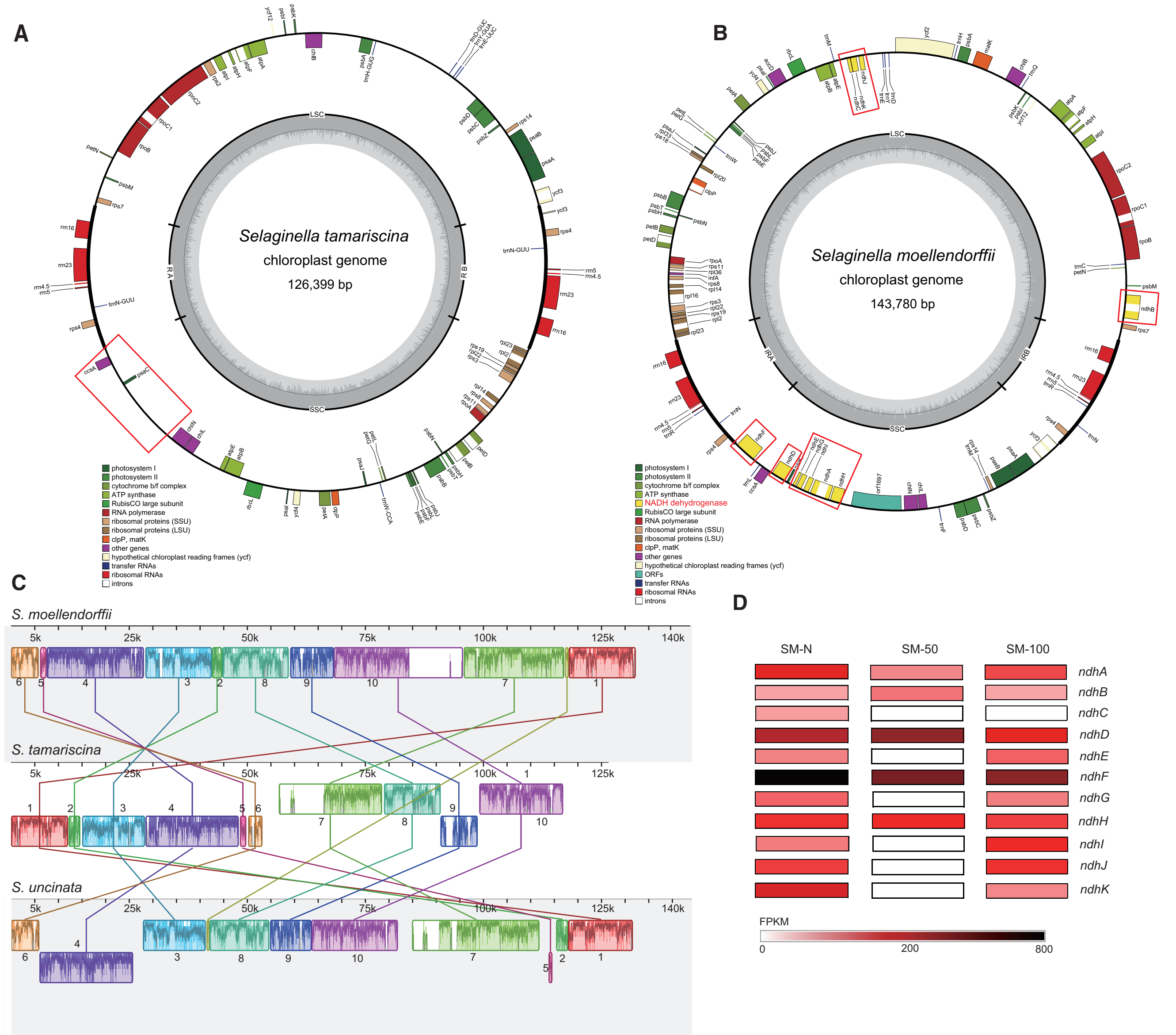

Figure 3. Gene Loss and Structural Rearrangement of the *S. tamariscina* Chloroplast Genome.
(A) The complete chloroplast genome of *S. tamariscina*.
(B) The complete chloroplast genome of *S. moellendorffii*. The red frames indicate the location of *cp-ndh* genes.
(C) Mauve alignment of the entire chloroplast genomes of *S. tamariscina*, *S. moellendorffii*, and *S. uncinata*.
(D) The differential expression (FPKM) of *cp-ndh* genes in *S. moellendorffii* under natural condition (SM-N), 50% RWC (SM-50, 1.02 g H_2O g^{-1} dwt), and 100% RWC (SM-100, 2.06 g H_2O g^{-1} dwt).

and *rps18*), and two hypothetical chloroplast open reading frames (*ycf1* and *ycf2*) were also missing. The *cp-ndhs*, *rpl20*, *rpl33*, *rps18*, *ycf1*, and *ycf2* genes were not transferred into the nuclear genome of *S. tamariscina*. In contrast, genes homologous to *rpl2* (ST78G561), *rpl16* (ST369G01), and *rpl36* (ST158G00) were detected in the nuclear genome.

The expression patterns of chloroplast genes in *S. tamariscina* and *S. moellendorffii* during drought were analyzed (Supplemental Tables 9 and 10). Under all conditions tested, the expression of 23.63% (13/55) and 21.25% (17/80) of the chloroplast genes in *S. tamariscina* and *S. moellendorffii*, respectively, showed no changes. Surprisingly, all *S. moellendorffii ndh* genes were expressed in the hydrated state, but no expression (6/11) or decreased expression (2/11) of most *ndh* genes was observed during dehydration.

Species-Specific Expansion and Contraction of Gene Families and Altered Expression of Genes Related to Resurrection Physiology

During dehydration, the aerial parts of *S. tamariscina* gradually curl into a brown ball and undergo chlorophyll degradation; then, after rehydration, the resurrection plant rapidly regains photosynthetic activity (Figure 1). To evaluate the gene expression changes of *S. tamariscina* plants under natural condition, 50% relative water content (RWC; 0.75 g H_2O g^{-1} dry weight [dwt]), and 100% RWC (1.49 g H_2O g^{-1} dwt), we performed the transcriptomic analyses and found that 1144 genes were significantly upregulated ($|\log_2$ ratio [50%RWC/natural]$| > 1$, $P < 0.05$) at 50% RWC. Functional annotation and gene ontology (GO) enrichment analysis of these genes were further performed to examine their potential relationships with the DT of *S. tamariscina*. The enriched GO categories were "transcription regulator," "calcium ion binding," and "response to stimulus," including genes such as protein phosphatase 2C (GO:0006464), abscisic acid (ABA)-insensitive protein (GO:0034641), and calcium-dependent protein kinase (GO:0016301) (Supplemental Figures 4 and 5). At 50% RWC, the expression of 5.87% (1219) of the genes in the *S. tamariscina* genome was significantly downregulated, including 242 unannotated genes (Supplemental Table 11). Among the downregulated genes, 325 genes were classified into primary energy metabolism categories related to cell metabolism, electron transfer, cellular protoplasm shrinkage, and cell wall remodeling, including genes belonging to the ATPase and ribonuclease families, ABC transporters, carbohydrate phosphorylases, and glyceraldehyde 3-phosphate dehydrogenases. This suggests that energy metabolism pathways are strongly repressed or deactivated as the RWC decreases.

We identified a total of 1305 PPR genes in the *S. tamariscina* genome, more than double those in *Glycine max* that was previously reported to have the most PPR genes among land plants, and nearly 3-fold those in *A. thaliana* (Fujii and Small, 2011). Interestingly, a similar number (1476) of PPR proteins were annotated in *S. moellendorffii*. Given these results, we scanned the available moss and algal genomes and found 464 PPR repeat genes in *Physcomitrella patens*, 21 PPRs in *Chlamydomonas reinhardtii*, and 31 PPRs in *Coccomyxa subellipsoidea*, suggesting a great expansion of PPR genes in lycophytes. A total of 83 PPR genes, which are required for chloroplast development in model plants (Ramos-Vega et al., 2015; Guillaumot et al., 2017), were downregulated at 50% RWC; this repression is likely related to chloroplast disassembly. We also found that the expression of 66 genes related to chlorophyll synthesis and photosynthesis was decreased at 50% RWC, indicating that chlorophyll degradation occurs during dehydration (Supplemental Table 11). These downregulated genes include 10 genes encoding chlorophyll *a–b* binding proteins (Cabs), which function as photochemical reaction centers in light absorption.

Distant relatives of the Cab genes, the early light-induced protein (ELIP) family members, mainly prevent oxidative damage by binding with the chlorophyll (Heddad et al., 2006). In the resurrection plants *Craterostigma plantagineum* (Bartels et al., 1992), *B. hygrometrica* (Xiao et al., 2015), and *S. lepidophylla* (VanBuren et al., 2018), the ELIP gene family has undergone dramatic expansion, and the expression of ELIP genes significantly increases in response to drought stress. However, in this study we found that the ELIP gene family has contracted in the resurrection plants *S. tamariscina*, *X. viscosa*, and *O. thomaeum*. Only one and two ELIP genes were identified in the *S. tamariscina* and *X. viscosa* genomes, respectively, which is similar to the number found in the genomes of many desiccation-sensitive plants, such as *S. moellendorffii*, *A. thaliana*, *Solanum tuberosum*, and *V. vinifera* (Supplemental Table 12). Interestingly, expression of the single ELIP gene in *S. tamariscina* was downregulated 10.5-fold (P = 1.1E−3) in response to 50% dehydration. Decreased expression of the ELIP gene in *S. moellendorffii* during dehydration was also observed. No ELIP gene was identified in the genome of *O. thomaeum*, indicating that protection against oxidative damage by ELIPs is absent in the desiccation-tolerant plants *S. tamariscina* and *O. thomaeum*.

Universal DT-related genes in seeds and vegetative tissues include the late embryogenesis abundant (LEA) genes, heat-shock protein genes, and ABA signaling pathway genes (Delahaie et al., 2013; Xiao et al., 2015; Giarola et al., 2017), all of which were highly expressed in *S. tamariscina* during dehydration (Supplemental Table 13). The *S. tamariscina* genome contains 40 LEA protein genes, which are classified into seven LEA subfamilies, namely LEA_1 (2), LEA_2 (14), LEA_3 (5), LEA_4 (6), LEA_5 (8), seed maturation protein (SMP) (3), and dehydrin (DHN) (2) (Supplemental Table 12). The number of LEA genes in the genomes of *B. hygrometrica* (55), *S. lepidophylla* (65), *O. thomaeum* (83), and *X. viscosa* (126) varies, suggesting no direct correlation between the number of LEA genes and DT. A significant expansion of the LEA_5 subfamily was observed in *S. tamariscina*. Differential expression analysis revealed that 12 LEA genes were significantly induced ($P < 0.05$) by 50% RWC in *S. tamariscina* (Supplemental Tables 14 and 15). Remarkably, the expression levels of four LEA genes increased 252.8-, 21.0-, 13.7-, and 12.8-fold after dehydration, with FPKM (fragments per kilobase of exon per million fragments mapped) values of >1000 (P = 5E−5, P = 5E−5, P = 2E−4, P = 4E−2), supporting the hypothesis that these genes function in the primary response to drought stress and regulation of homeostasis in desiccating cells. In contrast, the genome of the desiccation-sensitive *S. moellendorffii* contains 24 LEA genes, 15 of which were expressed and eight of which were differentially expressed in response to dehydration ($|\log_2$ ratio$| > 1$), suggesting a common induction of LEA genes in response to drought stress in land plants (Supplemental Table 14). However, the lack of a DHN gene in the *S. moellendorffii* genome, and extremely high expression of the DHN gene (FPKM = 14 044) at 50% RWC in *S. tamariscina* indicates the importance of DHN in the DT in *S. tamariscina*.

Lipid droplets are enclosed by phospholipids and oleosin store triacylglycerols that are used as a source of energy during the germination of plant seeds with desiccation-tolerant traits (Huang and Huang, 2015; VanBuren et al., 2017). Oleosin genes are conserved in land plants and absent in chlorophytes (Figure 4A and Supplemental Table 12). In seed plants, significant expansion of the oleosin gene family has been reported in *A. thaliana*, which contains 17 oleosin genes. However, an unexpected expansion of the oleosin gene family was also identified in the *S. tamariscina* lineage (Supplemental Table 16). A comparison of the genomic and phylogenetic relationships between oleosin genes from different species indicated that the oleosin genes in *S. tamariscina* (29 in total) could be divided into two groups, including 23 primitive (P) and six universal (U) oleosins. Seven oleosin genes were found in the genome of the desiccation-sensitive *S. moellendorffii*, which were constitutively expressed at a low level (Figure 4B).

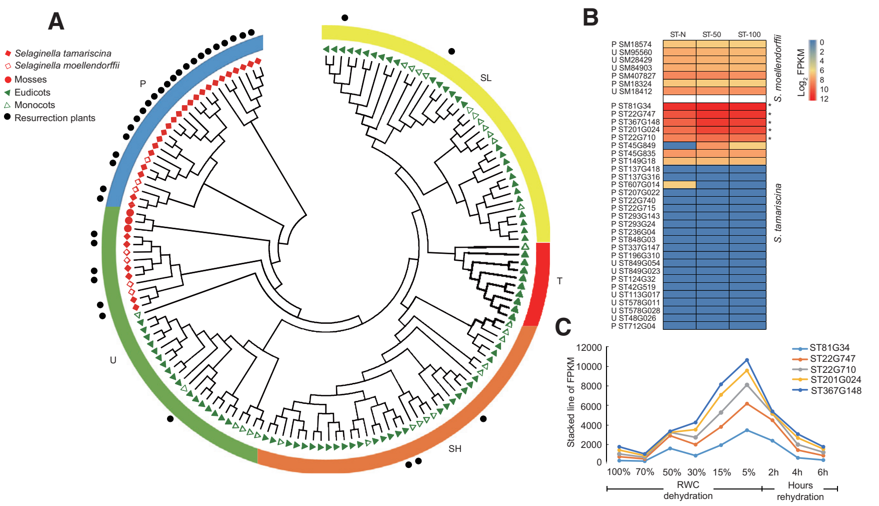

Figure 4. Phylogenetic Analysis and Differential Expression of the Oleosin Genes under Dehydration.

(A) Phylogenetic analysis of the oleosin gene family in *S. tamariscina* and 17 additional plant genomes.

(B) The heatmap represents gene expression levels (log_2 FPKM) under natural condition (ST-N), 50% RWC (ST-50), and 100% RWC (ST-100).

(C) The differential expression (FPKM) of five oleosin genes under HD (hydration), 70% RWC, 50% RWC, 30% RWC, 15% RWC, 5% RWC, RH 2 h (rehydration), RH 4 h, and RH 6 h.

Examination of gene expression in *S. tamariscina* under continuous dehydration and rehydration indicated that five oleosin genes were significantly upregulated during dehydration and downregulated during rehydration, with the highest expression levels observed at 5% RWC (0.07 g H2O g^{-1} dwt). The other 24 oleosin genes, including all oleosins from the P group, showed low or no expression (FPKM < 10) (Figure 4C and Supplemental Table 17). Interestingly, the expression of three oleosin genes (ST81G34, ST22G747, and ST367G148) represented 90.8% of all expressed oleosin transcripts.

Specific Regulation of ROS and the ABA Signaling Pathway during Drought Response

In plants, ROS significantly accumulate during abiotic stress responses, and ROS generation can cause oxidative damage and cell death (You and Chan, 2015). In *A. thaliana* and other model plants, several types of enzymes, such as NAD(P)H oxidases (NOX), amine oxidases (AOC), polyamine oxidases (PAO), and oxalate oxidases (OAO), contribute to ROS generation (Inupakutika et al., 2016).We identified 28 homologous genes encoding these enzymes in *S. tamariscina* by searching against *A. thaliana* sequences (Figure 5). By comparing the genomes of the 18 species, we found that respiratory burst oxidase homolog (RBOH) and OAO genes are absent in the hydrophyte chlorophytes, including *C. reinhardtii*, *C. subellipsoidea*, and *Dunaliella salina* (Figure 5 and Supplemental Table 18). During the evolution of land plants, the RBOH genes appeared in mosses and vascular plants. Notably, the number of RBOH, PAO, and OAO genes related to ROS generation in *S. moellendorffii* is twice that in *S. tamariscina* and *O. thomaeum*, suggesting less ROS production in these resurrection plants (Figure 5). Interestingly, no significant increase in the expression of the genes in the AOC, PAO, and OAO families occurred in *S. tamariscina* at 50% RWC.

Recent studies have revealed that ROS are also key signal transduction molecules in the regulation of many biological processes (Mignolet-Spruyt et al., 2016; Mittler, 2017), and balancing the ROS level is important for resisting and adapting to environmental stress (Miller, 2012; Noctor et al., 2014). A potentially strong ROS scavenging mechanism was identified in *S. tamariscina* but not in *S. moellendorffii* (Figure 6). Superoxide dismutase (SOD) acts as the first line of defense, converting $O_2^{\bullet-}$ into H_2O_2, after which catalase (CAT), ascorbate peroxidase (APX), and glutathione peroxidase (GPX) detoxify H_2O_2 (Figure 6 and Supplemental Table 18). Fe/Mn-SOD genes are conserved throughout the plant kingdom, from hydrophytes to terrestrial plants; however, CuZn-SOD genes are absent in chlorophytes. The Fe/Mn-SOD protein in chlorophytes is the ancestral form of plant SODs, which then evolved into distinct Fe-SOD and Mn-SOD groups in the moss and vascular plant lineages (Supplemental Figure 9). The genes encoding CuZn-SOD, which had been identified as the key ROS scavenging enzyme in response to drought stress (Miller, 2012), were expressed at an extremely high level during dehydration in *S. tamariscina* (Figure 6). Two genes (ST75G20 and ST134G15) encoding CAT were also highly expressed in *S. tamariscina*; for example, the FPKM value of ST75G20 increased from 2016 to 3359 in response to drought stress.

The ABA signaling pathway is central to drought stress response in seed plants; however, the role of ABA in the control of stomatal opening and closure in ferns and lycophytes is controversial (McAdam and Brodribb, 2012a, 2012b; ; McAdam et al., 2016). We identified 11 NCED (9-*cis*-epoxycarotenoid dioxygenases), five ABA1 (zeaxanthin epoxidase), eight ABA2 (short-chain dehydrogenases/reductases, SDR110C subfamily) (Supplemental Figure 10), and four ABA3 (aldehyde oxidases) genes in the *S. tamariscina* genome (Supplemental Table 19). In *S. tamariscina*, there is a significant increase in endogenous ABA content after dehydration (Wang et al., 2010). Consistently, we found that two ABA biosynthesis-related genes were significantly upregulated in *S. tamariscina* during dehydration: the expression level of one NCED gene was remarkably increased (121.8-fold, P = 3.0E−4), and that of one ABA3 gene was increased approximately 4.4-fold (P = 1.6E−3). Eight genes encoding putative PYR/PYL/RCAR (PYLs) ABA receptors were found in the *S. tamariscina* genome. In addition, 46 PP2C family members, classified into 11 clades, were identified. This is far fewer than the number of PP2C genes in *A. thaliana* (80) and *O. sativa* (78) (Xue et al., 2008) (Supplemental Figure 11). Members of the PP2CA clade have been described as co-receptors with the PYLs and negative regulators in the ABA signaling pathway (Zhu, 2016), which showed increased expression in response to ABA treatment and diverse environmental stress stimuli. Three PP2CA genes were found in

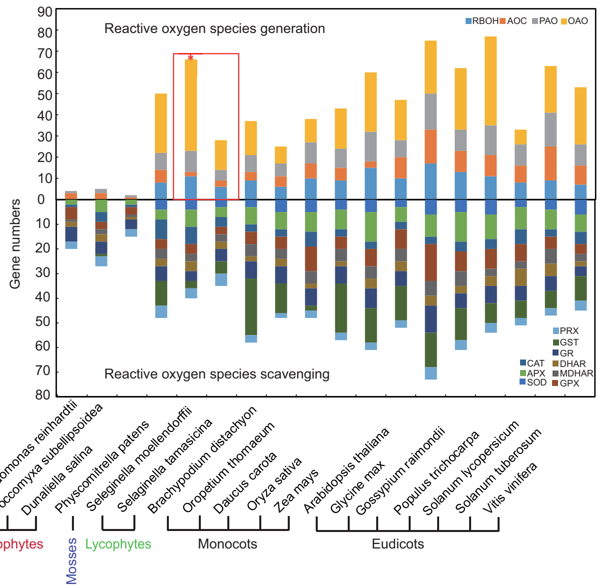

Figure 5. The Evolution of the ROS Generation and Scavenging Mechanism from Chlorophytes to Land Plants.

The different colors represent different ROS genes in 18 plant species. The red frame and asterisk show the significant differences in ROS generation gene numbers between *S. tamariscina* and *S. moellendorffii*.

S. tamariscina, and their expression levels increased significantly at 50% RWC, 15.4-fold ($P = 2.0E{-}4$), 14.2-fold ($P = 1.0E{-}3$), and 4.5-fold ($P = 6.3E{-}3$) elevation, respectively, suggesting a link between the negative regulation of ABA signaling and positive regulation of environmental stress tolerance. Two slow anion channel-associated genes, which regulate stomatal closure in seed plants, were identified in *S. tamariscina*; however, both genes showed extremely low expression (Figure 6).

DISCUSSION

This study presents the genome sequence of a medicinal resurrection plant assembled by SMRT sequencing. The *S. tamariscina* genome obtained in this study is a valuable resource for understanding the evolution of resurrection plants. The long reads obtained from third-generation sequencing have a significant advantage in the assembly of highly heterozygous genome such as *S. tamariscina* compared with the short reads generated by next-generation sequencing. The sequence of the *S. tamariscina* genome, together with the genome sequences of *S. moellendorffii*, mosses, algae, and several angiosperms, enable us to have a better understanding of the early evolution of land plants and how resurrection plants acquired the DT (Banks, 2009; Banks et al., 2011; Baniaga et al., 2016).

By comparing the genomes of desiccation-sensitive *S. moellendorffii* and desiccation-tolerant *S. tamariscina*, unique biological characteristics related to extreme environmental adaptations were revealed. The number of miRNAs and the proportion of repetitive DNA in *S. tamariscina* are significantly higher than those in *S. moellendorffii* (Banks et al., 2011), suggesting the potential regulation of miRNA and repetitive DNA during extreme drought conditions. Analysis of gene family evolution revealed sharp expansion of the PPR gene family in *S. tamariscina* and *S. moellendorffii*, implying that dramatic and lineage-specific PPR gene duplication occurred in lycophytes. The significant expansion and downregulated expression of the PPR gene family in *S. tamariscina* indicates their involvement in chloroplast development and disassembly in response to dehydration stress. The presence of distinct types of terpene synthases (TPS) genes in *S. moellendorffii* has raised the possibility of multiple evolutionary origins and horizontal gene transfer from microbes (Li et al., 2012); however, one-third of the *S. moellendorffii* TPS were identified in *S. tamariscina*, demonstrating that TPS evolution and plant–microbe symbiosis are dependent on the living environment. There is no significant variation in the expression of genes in the flavonoid biosynthesis pathway during dehydration, suggesting that the DT of *S. tamariscina* is not related to the biosynthesis of medicinal compounds during drought (Rafsanjani et al., 2015).

Gene loss and structural rearrangement of the chloroplast genome might have played an important role in the evolution of DT in *S. tamariscina*. Currently, more than 1670 land plant chloroplast genomes (1658 species, 794 genera) are available in the National Center for Biotechnology Information (NCBI) organelle genome database. Although chloroplast genomes have highly conserved structures, the size of these genomes vary, ranging from 19 kb in *Cytinus hypocistis*, a holoparasitic plant (Roquet et al., 2016), to 242 kb in *Pelargonium transvaalense*. Many studies have revealed the loss of genes or IR regions in certain lineages, such as the loss of the *cp-ndh* family in hemiparasitic, holoparasitic, and heterotrophic plants (Daniell et al., 2016; Li et al., 2017; Lin et al., 2017) and the loss of one copy of the IR in *Cassytha filiformis* and *Cassytha capillaris* (Song et al., 2017). The *cp-ndh* deletion in some orchids demonstrates that the NDH complex is not necessary for photoautotrophic plant species (Lin et al., 2017). Interestingly, the complete loss of *cp-ndh* genes in the desiccation-tolerant *S. tamariscina* was observed, which suggests that an unnecessary function related to photosynthesis was lost during the evolution of responses to rapid dehydration and resurrection traits. The terminated or decreased expression of most *cp-ndh* genes during dehydration in desiccation-sensitive *S. moellendorffii* provides evidence for a link between *cp-ndh* loss and resurrection in desiccation-tolerant *S. tamariscina*. Chloroplast IR regions have been described as evolutionary markers because many rearrangements have occurred at the boundaries of IR regions (Zhu et al., 2016). Unique forward repeat regions, named RA and RB, which are distinct from common IR regions, were detected in *S. tamariscina* (Figure 3A). In addition, the structural rearrangement of large fragments at the boundaries of RA and RB distinct from those in *S. moellendorffii* and *S. uncinata* was observed in *S. tamariscina*. Our findings indicate that the rearrangement of classic structures and the complete loss of a gene family have no influence on the photosynthesis system, suggesting that it will be possible to reduce and modify the chloroplast genome as part of efforts to create synthetic chloroplast factories.

The restructuring of complex gene regulatory networks, involving the repression or activation of many genes in response to dehydration, plays an essential role in the acquisition of DT in resurrection plants (Gechev et al., 2012; Giarola et al., 2017). Comparisons of the genome of *S. tamariscina* and other

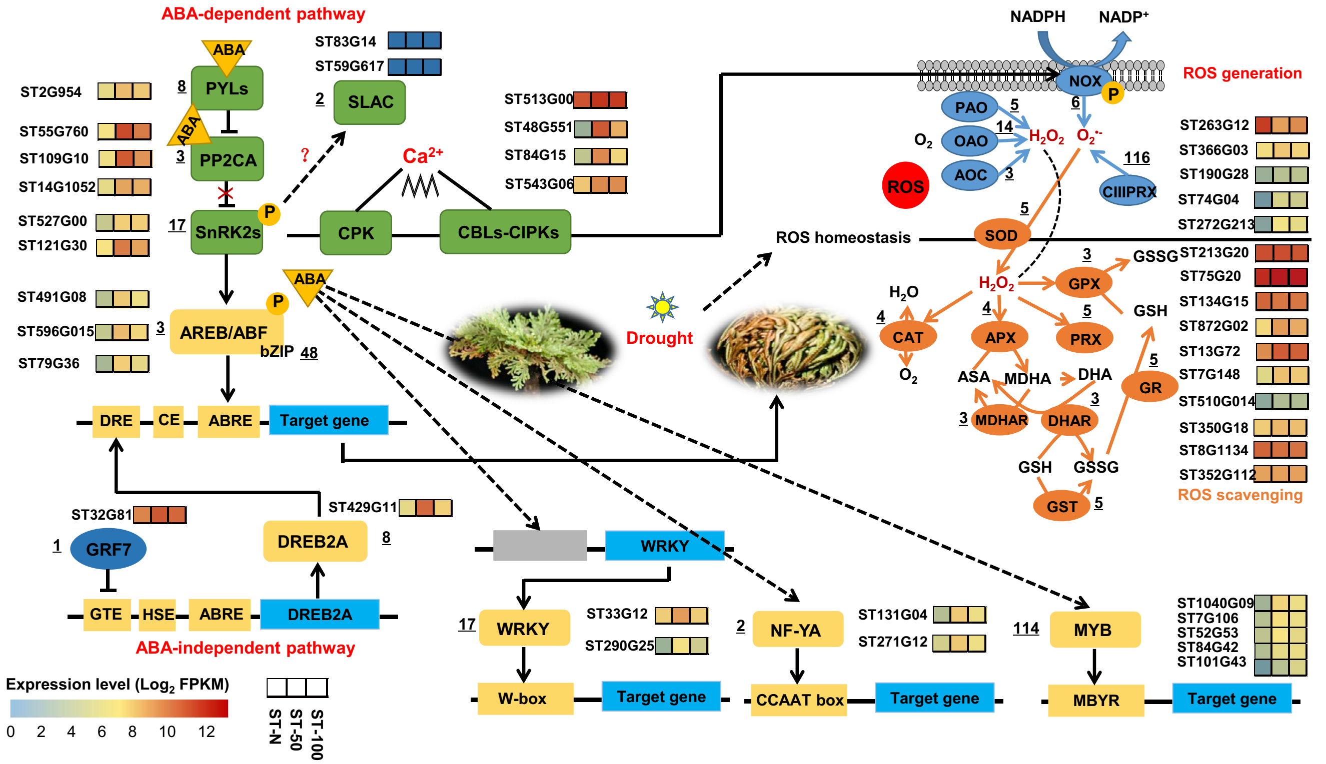

Figure 6. A Blueprint of the Resurrection Mechanism in *S. tamariscina*, including the ABA-Dependent Pathway, the ABA-Independent Pathway, and ROS Homeostasis, Determined via Comparative Genomics.

The heatmap shows gene expression levels (log_2 FPKM) under natural condition (ST-N), 50% RWC (ST-50), and 100% RWC (ST-100). The underlined numbers indicate the number of protein-coding genes in the *S. tamariscina* genome.

resurrection plants can potentially reveal species-specific resurrection mechanisms. The expansion and extremely high abundance of the P group oleosin genes, which are related to energy storage in plant seeds, reflects their ancestral evolution and protective function in the DT mechanism of *S. tamariscina*. Oleosins, which serve as high-density energy sources and function in membrane repair during desiccation, could prevent the coalescence of oil bodies and protect membrane integrity. The protective mechanism of oleosins in resurrection plants might be a manifestation of the recapitulation of seed developmental reprogramming in vegetative tissues, which is seen in angiosperms. The more pronounced contraction of ELIP genes in *S. tamariscina* compared with *S. lepidophylla* reveals a major difference in the defense mechanism against drought between these species. The expansion and contraction of the LEA gene family in different species indicates a negative correlation with DT traits; however, the significant upregulation of LEA genes in *S. tamariscina* suggests that they provide important protection during dehydration.

The genomic and transcriptomic data from resurrection plants provide a wealth of information that can be used to study the conserved mechanisms of DT. For example, through comparative genomic and transcriptomic analysis, we derived a blueprint of the DT mechanism in *S. tamariscina*. The genome of *S. tamariscina* will also be valuable for unraveling the acquisition of common or species-specific DT traits and provide important implications for guiding drought tolerance improvement of crops via different strategies.

METHODS

Sample Collection and Genome Survey

S. tamariscina plants were collected from Lishui City (28°N, 118°E), Zhejiang Province, China. *S. tamariscina* was identified using DNA barcoding technology (Gu et al., 2013) and by the morphologist Prof. Yulin Lin from the Institute of Medicinal Plant Development. Fifty micrograms of high-molecular-weight genomic DNA was extracted following megabase-size DNA preparation methods (Zhang et al., 2012). Two libraries with DNA insert sizes of 250 and 500 bp were constructed from *S. tamariscina* genomic DNA, and 2 × 125 paired-end sequencing was performed using the Illumina HiSeq 4000 platform. The *k*-mer distribution of Illumina reads was analyzed to estimate the genome size of *S. tamariscina*. The genome was primarily assembled using SOAPdenovo2 software (Luo et al., 2012).

Dehydration Stress in *S. tamariscina* and *S. moellendorffii* Plants

Plants received three different treatments: natural condition (representing the habitat of *S. tamariscina* and *S. moellendorfii*), drought, and rehydration. For the drought treatment, the plants were placed on dry trays in a greenhouse. Six plants of each species were air-dried to 50% RWC. For the rehydration treatment, three plants of each species were then rehydrated for 24 h in distilled water to 100% RWC. For the natural treatment, three plants of each species were grown in an environment representing the natural, moist habitat of *S. tamariscina* and *S. moellendorfii*. RWC was calculated using the formula RWC (%) = (F_{wt} − D_{wt})/(FT_{wt} − D_{wt}) × 100 (Yobi et al., 2012), where F_{wt} is the fresh weight after dehydration for a specific amount of time, D_{wt} represents the dry weight after incubation at 65°C for 2 days, and FT_{wt} is the fresh weight after rehydration for 24 h. The total water content (TWC) was also calculated using the formula

TWC = (F_{wt} − D_{wt})/D_{wt} (Costa et al., 2017). In addition, a continuous dehydration and rehydration experiment was performed, whereby the water content of plants changed as follows: HD (hydration), 70% RWC, 50% RWC, 30% RWC, 15% RWC, 5% RWC (2 weeks without water), RH 2 h (rehydration), RH 4 h, and RH 6 h. Photosynthetic activity of *S. tamariscina* under dehydration and rehydration was recorded using MAXI IMAGING-PAM (Walz, Germany) (Oliver and Erhard, 2008).

SMRT Sequencing and Genome Assembly

Twenty-kilobase DNA templates were prepared using the BluePippin size-selection system (PAC20KB/BLF7510). The templates were bound to P6 DNA polymerase and V2 primers using the DNA/polymerase Binding Kit P6 v2. *S. tamariscina* genomic DNA was sequenced using 20 SMRT cells on the PacBio RS II platform. The raw data were filtered to remove adapters and low-quality reads using SMRT Portal.

The PacBio raw reads were corrected, trimmed, and assembled into genomic contigs using FALCON (https://github.com/PacificBiosciences/FALCON). The length cutoff was set at 5000 bp for the initial mapping, preassembly, and error correction. The minimum and maximum coverage values were set to 2 and 100 for overlap detection and filtering of error-corrected reads. The assembled contigs were used to construct scaffolds using SSPACE-LongRead (version 1.1). Illumina sequences from the *S. tamariscina* DNA and RNA libraries were mapped to evaluate the quality of the assembled genome.

Genome Annotation

Structural repeat annotation of the *S. tamariscina* genome was performed using RepeatModeler (http://www.repeatmasker.org/RepeatModeler/; version 1.0.9) software. Two *de novo* repeat finding programs, RECON and RepeatScout, were employed for identifying and classifying repeat elements. Repbase (version 21.12) was also used, and consensus classification of transposable elements (TEs) was performed according to the default parameters. Finally, RepeatMasker (version 4.0.6) was used to count and mask the TE sequences. RMBlast, a modified version of the NCBI BLAST program, was used for all-by-all alignments done with RepeatModeler and RepeatMasker.

High-quality RNA was separately isolated and sequenced from control samples (natural condition), dehydrated samples (50% RWC), and rehydrated samples (100% RWC) of *S. tamariscina* and *S. moellendorffii*. *De novo* transcript assembly was done using Trinity (Grabherr et al., 2011) (version 2.2.0) with the default parameters, and peptide sequences were predicted by TransDecoder (https://github.com/TransDecoder; version 2.1.0). *Ab initio* predictions of protein-coding genes were done using the MAKER annotation pipeline (version 2.31.9) (Campbell et al., 2014; Cantarel et al., 2008), by integrating the assembled transcripts of *S. tamariscina* and protein sequences from *S. moellendorffii*, *A. thaliana*, and *S. tamariscina*.

Comparative Genomic and Transcriptomic Analyses

OrthoMCL (Li et al., 2003) (version 2.0.9) was used to cluster gene families from *S. tamariscina* and 17 other vascular and non-vascular plants. Then, all_vs_all alignments were performed using BLASTP with an E-value cutoff of 1E−5. The Markov cluster algorithm (MCL) was used to determine species-specific gene groups from similar matrices. CAFÉ (De Bie et al., 2006) (version 3.1) was used to predict gene family expansion and contraction. Phylogenetic analysis of these 18 species was performed with 18 high-quality 1:1 single-copy orthologous genes using the RAxML package (Stamatakis, 2006) (version 8.1.13). Divergence times were directly estimated based on the divergence times of *Populus trichocarpa*–*G. max* (94–127 MYA), *A. thaliana*–*Gossypium raimondii* (81–94 MYA), and *S. tuberosum*–*Solanum lycopersicum* (5.09–10.25 MYA) obtained from TimeTree (http://www.timetree.org).

The transcriptomes of *S. tamariscina* and *S. moellendorffii* under natural condition, 50% RWC, and 100% RWC were sequenced using the Illumina HiSeq 2500 platform. For the acquirement of additional information, the following nine transcriptomes of *S. tamariscina* under different hydration conditions were also sequenced: HD (hydration), 70% RWC, 50% RWC, 30% RWC, 15% RWC, 5% RWC, RH 2 h (rehydration), RH 4 h, and RH 6 h. The raw data were cleaned and mapped to the assembled *S. tamariscina* genome and the published *S. moellendorffii* genome using Trimmomatic (version 0.30) and HISAT2 (version 2.0.5). Cufflinks (Trapnell et al., 2012) (version 2.2.1) packages were then used to compute FPKM values and to determine the statistical significance of differential expression under dehydration. GO and KEGG enrichment analyses were performed to identify the biological processes, molecular functions, cellular components, and pathway maps. Differential expression was visualized using the "gplots" package of R.

Identification of DT-Related Proteins and Phylogenetic Analysis

For identification of the members of DT-related protein families, the following family hidden Markov models (HMMs) were downloaded from the Pfam database (http://pfam.xfam.org/): LEA (LEA_1, PF03760; LEA_2, PF03168; LEA_3, PF03242; LEA_4, PF02987; LEA_5, PF00477; LEA_6, PF10714; DHN, PF00257; SMP, PF04927), oleosin (PF01277), and caleosin (PF05042). The HMM matrix was generated and scanned using the hmmpress and hmmscan scripts from the HMMER package (version 3.1b2) with a domain E-value threshold of 1E−10. Full-length amino acid sequences were aligned using the ClustalW2 program in MEGA (version 7). Phylogenetic trees were constructed using the neighbor-joining method with the Jones–Taylor–Thornton model and 1000 bootstrap replicates.

ACCESSION NUMBERS

Genome sequences, transcriptome sequences, and assembly data were submitted to the Sequence Read Archive from the NCBI under the accession numbers SAMN07840812 (BioSample ID) and PRJNA416064 (BioProject ID).

SUPPLEMENTAL INFORMATION

Supplemental Information is available at *Molecular Plant Online*.

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AUTHOR CONTRIBUTIONS

Conceptualization, Z.X., J.S., and S.C.; Methodology, Z.X., T.X., Y.L., W.G., H. Yao, H. Yu, X.P., and C.X.; Investigation, Z.X., Y.L., and T.X.; Formal Analysis, Z.X., Y.L., J.Z., S.L., J.X., and J.S.; Writing – Original Draft, Z.X. and J.S.; Writing – Review & Editing, S.C., H.L., and D.B.

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REFERENCES

Baniaga, A.E., Arrigo, N., and Barker, M.S. (2016). The small nuclear genomes of *Selaginella* are associated with a low rate of genome size evolution. Genome Biol. Evol. **8**:1516–1525.

Banks, J.A. (2009). *Selaginella* and 400 million years of separation. Annu. Rev. Plant Biol. **60**:223–238.

Banks, J.A., Nishiyama, T., Hasebe, M., Bowman, J.L., Gribskov, M., dePamphilis, C., Albert, V.A., Aono, N., Aoyama, T., Ambrose, B.A., et al. (2011). The *Selaginella* genome identifies genetic changes associated with the evolution of vascular plants. Science **332**:960–963.

Bartels, D., Hanke, C., Schneider, K., Michel, D., and Salamini, F. (1992). A desiccation-related Elip-like gene from the resurrection plant *Craterostigma plantagineum* is regulated by light and ABA. EMBO J. **11**:2771–2778.

Campbell, M.S., Holt, C., Moore, B., and Yandell, M. (2014). Genome annotation and curation using MAKER and MAKER-P. Curr. Protoc. Bioinformatics **48**:11–39.

Cantarel, B.L., Korf, I., Robb, S.M., Parra, G., Ross, E., Moore, B., Holt, C., Sánchez Alvarado, A., and Yandell, M. (2008). MAKER: an easy-to-use annotation pipeline designed for emerging model organism genomes. Genome Res. **18**:188–196.

Chen, S.L., Song, J.Y., Sun, C., Xu, J., Zhu, Y.J., Verpoorte, R., and Fan, T.P. (2015). Herbal genomics: examining the biology of traditional medicines. Science **347** (6219 Suppl):S27–S29.

Chinese Pharmacopoeia Committee. (2015). Pharmacopoeia of the People's Republic of China (Beijing, China: China Medical Science and Technology Press).

Costa, M.C., Farrant, J.M., Oliver, M.J., Ligterink, W., Buitink, J., and Hilhorst, H.M. (2016). Key genes involved in desiccation tolerance and dormancy across life forms. Plant Sci. **251**:162–168.

Costa, M.D., Artur, M.A., Maia, J., Jonkheer, E., Derks, M.F., Nijveen, H., Williams, B., Mundree, S.G., Jimenez-Gomez, J.M., Hesselink, T., et al. (2017). A footprint of desiccation tolerance in the genome of *Xerophyta viscosa*. Nat. Plants **3**:17038.

Daniell, H., Lin, C.S., Yu, M., and Chang, W.J. (2016). Chloroplast genomes: diversity, evolution, and applications in genetic engineering. Genome Biol. **17**:134.

De Bie, T., Cristianini, N., Demuth, J.P., and Hahn, M.W. (2006). CAFE: a computational tool for the study of gene family evolution. Bioinformatics **22**:1269–1271.

Deeba, F., Pandey, A.K., and Pandey, V. (2016). Organ specific proteomic dissection of *Selaginella bryopteris* undergoing dehydration and rehydration. Front. Plant Sci. **7**:425.

Delahaie, J., Hundertmark, M., Bove, J., Leprince, O., Rogniaux, H., and Buitink, J. (2013). LEA polypeptide profiling of recalcitrant and orthodox legume seeds reveals ABI3-regulated LEA protein abundance linked to desiccation tolerance. J. Exp. Bot. **64**:4559–4573.

Duff, R.J., and Nickrent, D.L. (1999). Phylogenetic relationships of land plants using mitochondrial small-subunit rDNA sequences. Am. J. Bot. **86**:372–386.

Farrant, J.M., and Moore, J.P. (2011). Programming desiccation-tolerance: from plants to seeds to resurrection plants. Curr. Opin. Plant Biol. **14**:340–345.

Fujii, S., and Small, I. (2011). The evolution of RNA editing and pentatricopeptide repeat genes. New Phytol. **191**:37–47.

Gaff, D.F., and Oliver, M. (2013). The evolution of desiccation tolerance in angiosperm plants: a rare yet common phenomenon. Funct. Plant Biol. **40**:315–328.

Gechev, T.S., Dinakar, C., Benina, M., Toneva, V., and Bartels, D. (2012). Molecular mechanisms of desiccation tolerance in resurrection plants. Cell Mol. Life Sci. **69**:3175–3186.

Giarola, V., Hou, Q., and Bartels, D. (2017). Angiosperm plant desiccation tolerance: hints from transcriptomics and genome sequencing. Trends Plant Sci. **22**:705–717.

Grabherr, M.G., Haas, B.J., Yassour, M., Levin, J.Z., Thompson, D.A., Amit, I., Adiconis, X., Fan, L., Raychowdhury, R., Zeng, Q., et al. (2011). Full-length transcriptome assembly from RNA-Seq data without a reference genome. Nat. Biotechnol. **29**:644–652.

Gu, W., Song, J., Cao, Y., Sun, Q., Yao, H., Wu, Q., Chao, J., Zhou, J., Xue, W., and Duan, J. (2013). Application of the ITS2 region for barcoding medicinal plants of Selaginellaceae in pteridophyta. PloS One **8**:e67818.

Guillaumot, D., Lopez-Obando, M., Baudry, K., Avon, A., Rigaill, G., Falcon de Longevialle, A., Broche, B., Takenaka, M., Berthome, R., De Jaeger, G., et al. (2017). Two interacting PPR proteins are major *Arabidopsis* editing factors in plastid and mitochondria. Proc. Natl. Acad. Sci. USA **114**:8877–8882.

Heddad, M., Noren, H., Reiser, V., Dunaeva, M., Andersson, B., and Adamska, I. (2006). Differential expression and localization of early light-induced proteins in *Arabidopsis*. Plant Physiol. **142**:75–87.

Huang, M.D., and Huang, A.H. (2015). Bioinformatics reveal five lineages of oleosins and the mechanism of lineage evolution related to structure/function from green algae to seed plants. Plant Physiol. **169**:453–470.

Inupakutika, M.A., Sengupta, S., Devireddy, A.R., Azad, R.K., and Mittler, R. (2016). The evolution of reactive oxygen species metabolism. J. Exp. Bot. **67**:5933–5943.

Jiao, Y., Wickett, N.J., Ayyampalayam, S., Chanderbali, A.S., Landherr, L., Ralph, P.E., Tomsho, L.P., Hu, Y., Liang, H., Soltis, P.S., et al. (2011). Ancestral polyploidy in seed plants and angiosperms. Nature **473**:97–100.

Korall, P., and Kenrick, P. (2002). Phylogenetic relationships in Selaginellaceae based on rbcL sequences. Am. J. Bot. **89**:506–517.

Korall, P., and Kenrick, P. (2004). The phylogenetic history of Selaginellaceae based on DNA sequences from the plastid and nucleus: extreme substitution rates and rate heterogeneity. Mol. Phylogenet. Evol. **31**:852–864.

Leprince, O., and Buitink, J. (2015). Introduction to desiccation biology: from old borders to new frontiers. Planta **242**:369–378.

Li, L., Stoeckert, C.J., Jr., and Roos, D.S. (2003). OrthoMCL: identification of ortholog groups for eukaryotic genomes. Genome Res. **13**:2178–2189.

Li, G., Köllner, T.G., Yin, Y., Jiang, Y., Chen, H., Xu, Y., Gershenzon, J., Pichersky, E., and Chen, F. (2012). Nonseed plant *Selaginella moellendorffii* has both seed plant and microbial types of terpene synthases. Proc. Natl. Acad. Sci. USA **109**:14711–14715.

Li, Y., Zhou, J.G., Chen, X.L., Cui, Y.X., Xu, Z.C., Li, Y.H., Song, J.Y., Duan, B.Z., and Yao, H. (2017). Gene losses and partial deletion of small single-copy regions of the chloroplast genomes of two hemiparasitic *Taxillus* species. Sci. Rep. **7**:12834.

Lin, C.S., Chen, J.J.W., Chiu, C.C., Hsiao, H.C.W., Yang, C.J., Jin, X.H., Leebens-Mack, J., de Pamphilis, C.W., Huang, Y.T., Yang, L.H., et al. (2017). Concomitant loss of NDH complex-related genes within chloroplast and nuclear genomes in some orchids. Plant J. **90**:994–1006.

Luo, R., Liu, B., Xie, Y., Li, Z., Huang, W., Yuan, J., He, G., Chen, Y., Pan, Q., Liu, Y., et al. (2012). SOAPdenovo2: an empirically improved memory-efficient short-read *de novo* assembler. GigaScience **1**:18.

McAdam, S.A., and Brodribb, T.J. (2012a). Stomatal innovation and the rise of seed plants. Ecol. Lett. **15**:1–8.

McAdam, S.A., and Brodribb, T.J. (2012b). Fern and lycophyte guard cells do not respond to endogenous abscisic acid. Plant Cell **24**:1510–1521.

McAdam, S.A., Brodribb, T.J., Banks, J.A., Hedrich, R., Atallah, N.M., Cai, C., Geringer, M.A., Lind, C., Nichols, D.S., Stachowski, K., et al. (2016). Abscisic acid controlled sex before transpiration in vascular plants. Proc. Natl. Acad. Sci. USA **113**:12862–12867.

Mignolet-Spruyt, L., Xu, E., Idanheimo, N., Hoeberichts, F.A., Muhlenbock, P., Brosche, M., Van Breusegem, F., and Kangasjarvi, J. (2016). Spreading the news: subcellular and organellar reactive oxygen species production and signalling. J. Exp. Bot. **67**:3831–3844.

Miller, A.F. (2012). Superoxide dismutases: ancient enzymes and new insights. FEBS Lett. **586**:585–595.

Mittler, R. (2017). ROS are good. Trends Plant Sci. **22**:11–19.

Noctor, G., Mhamdi, A., and Foyer, C.H. (2014). The roles of reactive oxygen metabolism in drought: not so cut and dried. Plant Physiol. **164**:1636–1648.

Nordberg, H., Cantor, M., Dusheyko, S., Hua, S., Poliakov, A., Shabalov, I., Smirnova, T., Grigoriev, I.V., and Dubchak, I. (2014). The genome portal of the Department of Energy Joint Genome Institute: 2014 updates. Nucleic Acids Res. **42**:D26–D31.

Oliver, M., and Erhard, P. (2008). Photosynthesis in ripe strawberries (*Fragaria* × *ananassa*) recording by a MAXI IMAGING-PAM. PAM Appl. Notes **1**:19–20.

Oliver, M.J., Tuba, Z., and Mishler, B.D. (2000). The evolution of vegetative desiccation tolerance in land plants. Plant Ecol. **151**:85–100.

Pampurova, S., Verschooten, K., Avonce, N., and Van Dijck, P. (2014). Functional screening of a cDNA library from the desiccation-tolerant plant *Selaginella lepidophylla* in yeast mutants identifies trehalose biosynthesis genes of plant and microbial origin. J. Plant Res. **127**:803–813.

Porembski, S., and Barthlott, W. (2000). Granitic and gneissic outcrops (inselbergs) as centers of diversity for desiccation-tolerant vascular plants. Plant Ecol. **151**:19–28.

Rafsanjani, A., Brule, V., Western, T.L., and Pasini, D. (2015). Hydro-responsive curling of the resurrection plant *Selaginella lepidophylla*. Sci. Rep. **5**:8064.

Ramos-Vega, M., Guevara-Garcia, A., Llamas, E., Sanchez-Leon, N., Olmedo-Monfil, V., Vielle-Calzada, J.P., and Leon, P. (2015). Functional analysis of the *Arabidopsis thaliana* CHLOROPLAST BIOGENESIS 19 pentatricopeptide repeat editing protein. New Phytol. **208**:430–441.

Raubeson, L.A., and Jansen, R.K. (1992). Chloroplast DNA evidence on the ancient evolutionary split in vascular land plants. Science **255**:1697–1699.

Roquet, C., Coissac, E., Cruaud, C., Boleda, M., Boyer, F., Alberti, A., Gielly, L., Taberlet, P., Thuiller, W., Van Es, J., et al. (2016). Understanding the evolution of holoparasitic plants: the complete plastid genome of the holoparasite *Cytinus hypocistis* (Cytinaceae). Ann. Bot. **118**:885–896.

Song, Y., Yu, W.B., Tan, Y., Liu, B., Yao, X., Jin, J., Padmanaba, M., Yang, J.B., and Corlett, R.T. (2017). Evolutionary comparisons of the chloroplast genome in Lauraceae and insights into loss events in the *Magnoliids*. Genome Biol. Evol. **9**:2354–2364.

Stamatakis, A. (2006). RAxML-VI-HPC: maximum likelihood-based phylogenetic analyses with thousands of taxa and mixed models. Bioinformatics **22**:2688–2690.

Tautz, D., and Domazet-Loso, T. (2011). The evolutionary origin of orphan genes. Nat. Rev. Genet. **12**:692–702.

Trapnell, C., Roberts, A., Goff, L., Pertea, G., Kim, D., Kelley, D.R., Pimentel, H., Salzberg, S.L., Rinn, J.L., and Pachter, L. (2012). Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks. Nat. Protoc. **7**:562–578.

VanBuren, R., Bryant, D., Edger, P.P., Tang, H., Burgess, D., Challabathula, D., Spittle, K., Hall, R., Gu, J., Lyons, E., et al. (2015). Single-molecule sequencing of the desiccation-tolerant grass *Oropetium thomaeum*. Nature **527**:508–511.

VanBuren, R., Wai, C.M., Zhang, Q., Song, X., Edger, P.P., Bryant, D., Michael, T.P., Mockler, T.C., and Bartels, D. (2017). Seed desiccation mechanisms co-opted for vegetative desiccation in the resurrection grass *Oropetium thomaeum*. Plant Cell Environ. **40**:2292–2306.

VanBuren, R., Wai, C.M., Ou, S., Pardo, J., Bryant, D., Jiang, N., Mockler, T.C., Edger, P., and Michael, T.P. (2018). Extreme haplotype variation in the desiccation-tolerant clubmoss *Selaginella lepidophylla*. Nat. Commun. **9**:13.

Wang, X., Chen, S., Zhang, H., Shi, L., Cao, F., Guo, L., Xie, Y., Wang, T., Yan, X., and Dai, S. (2010). Desiccation tolerance mechanism in resurrection fern-ally *Selaginella tamariscina* revealed by physiological and proteomic analysis. J. Proteome Res. **9**:6561–6577.

Xiao, L., Yang, G., Zhang, L., Yang, X., Zhao, S., Ji, Z., Zhou, Q., Hu, M., Wang, Y., Chen, M., et al. (2015). The resurrection genome of *Boea hygrometrica*: a blueprint for survival of dehydration. Proc. Natl. Acad. Sci. USA **112**:5833–5837.

Xue, T., Wang, D., Zhang, S., Ehlting, J., Ni, F., Jakab, S., Zheng, C., and Zhong, Y. (2008). Genome-wide and expression analysis of protein phosphatase 2C in rice and *Arabidopsis*. BMC Genomics **9**:550.

Yobi, A., Wone, B.W., Xu, W., Alexander, D.C., Guo, L., Ryals, J.A., Oliver, M.J., and Cushman, J.C. (2012). Comparative metabolic profiling between desiccation-sensitive and desiccation-tolerant species of *Selaginella* reveals insights into the resurrection trait. Plant J. **72**:983–999.

Yobi, A., Wone, B.W., Xu, W., Alexander, D.C., Guo, L., Ryals, J.A., Oliver, M.J., and Cushman, J.C. (2013). Metabolomic profiling in *Selaginella lepidophylla* at various hydration states provides new insights into the mechanistic basis of desiccation tolerance. Mol. Plant **6**:369–385.

You, J., and Chan, Z. (2015). ROS regulation during abiotic stress responses in crop plants. Front. Plant Sci. **6**:1092.

Zhang, M., Zhang, Y., Scheuring, C.F., Wu, C.C., Dong, J.J., and Zhang, H.B. (2012). Preparation of megabase-sized DNA from a variety of organisms using the nuclei method for advanced genomics research. Nat. Protoc. **7**:467–478.

Zhu, J.K. (2016). Abiotic stress signaling and responses in plants. Cell **167**:313–324.

Zhu, A., Guo, W., Gupta, S., Fan, W., and Mower, J.P. (2016). Evolutionary dynamics of the plastid inverted repeat: the effects of expansion, contraction, and loss on substitution rates. New Phytol. **209**:1747–1756.